

OPTIMIZER-INDUCED FEATURE GEOMETRY SHAPES POST-HOC OOD DETECTION RELIABILITY

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Abstract

ID accuracy는 정답률을 요약하지만, confidence calibration과 distribution-shift robustness를 보장하지 않는다. 우리는 CIFAR-10 WRN-28-10을 SGD, Adam, AdamW와 LR-weight decay grid로 학습하고, ID validation accuracy만으로 선택한 후보들의 calibration, post-hoc OOD detection, feature geometry를 진단한다. Seed0 diagnostic draft는 비슷한 accuracy band 안에서도 raw feature OOD reliability가 갈라질 수 있으며, L2-normalized controls가 그 차이를 feature norm/covariance-scale geometry와 연결해 볼 수 있음을 시사한다.

Introduction: Why Accuracy Is Not Reliability

Accuracy summarizes correctness; reliability asks more.

- ID accuracy는 모델이 test distribution에서 정답을 얼마나 자주 맞히는지를 요약한다.
- 실제 배포 reliability는 confidence가 믿을 만한지, distribution-shift input을 정상 입력처럼 받아들이지 않는지까지 함께 확인해야 한다.
- 비슷한 ID accuracy라도 optimizer와 LR-weight decay 선택이 penultimate feature geometry를 다르게 만들면 post-hoc detector의 score landscape도 달라질 수 있다.
- 따라서 optimizer를 단순한 training speed 선택이 아니라 calibration과 feature-based OOD reliability를 바꿀 수 있는 실험 요인으로 다룬다.

Key Question

비슷한 Accuracy라도 optimizer가 만든 feature geometry가 모델의 신뢰성에 어떤 영향을 미치는가?

Reliability Failure: Miscalibration and OOD Acceptance

신뢰성 실패는 높은 confidence를 믿고 넘기거나 OOD 입력을 정상 입력처럼 넘기는 순간 downstream decision을 혼든다.

- Confidence는 보통 모델이 선택한 class의 softmax probability이고, actual accuracy는 비슷한 confidence bin 안에서 실제로 정답이었던 비율이다.
- Confidence와 actual accuracy가 크게 다르면 miscalibration이며, confidence-threshold 기반 승인/보류 결정이 불안정해진다.
- Post-hoc OOD detector는 score $s(x)$ 로 ID sample에는 높은 ID-like score를, OOD sample에는 낮은 score를 주어야 한다.
- OOD sample이 높은 ID-like score를 받으면, 지원하지 않는 입력도 정상 class prediction으로 downstream decision에 넘어갈 수 있다.

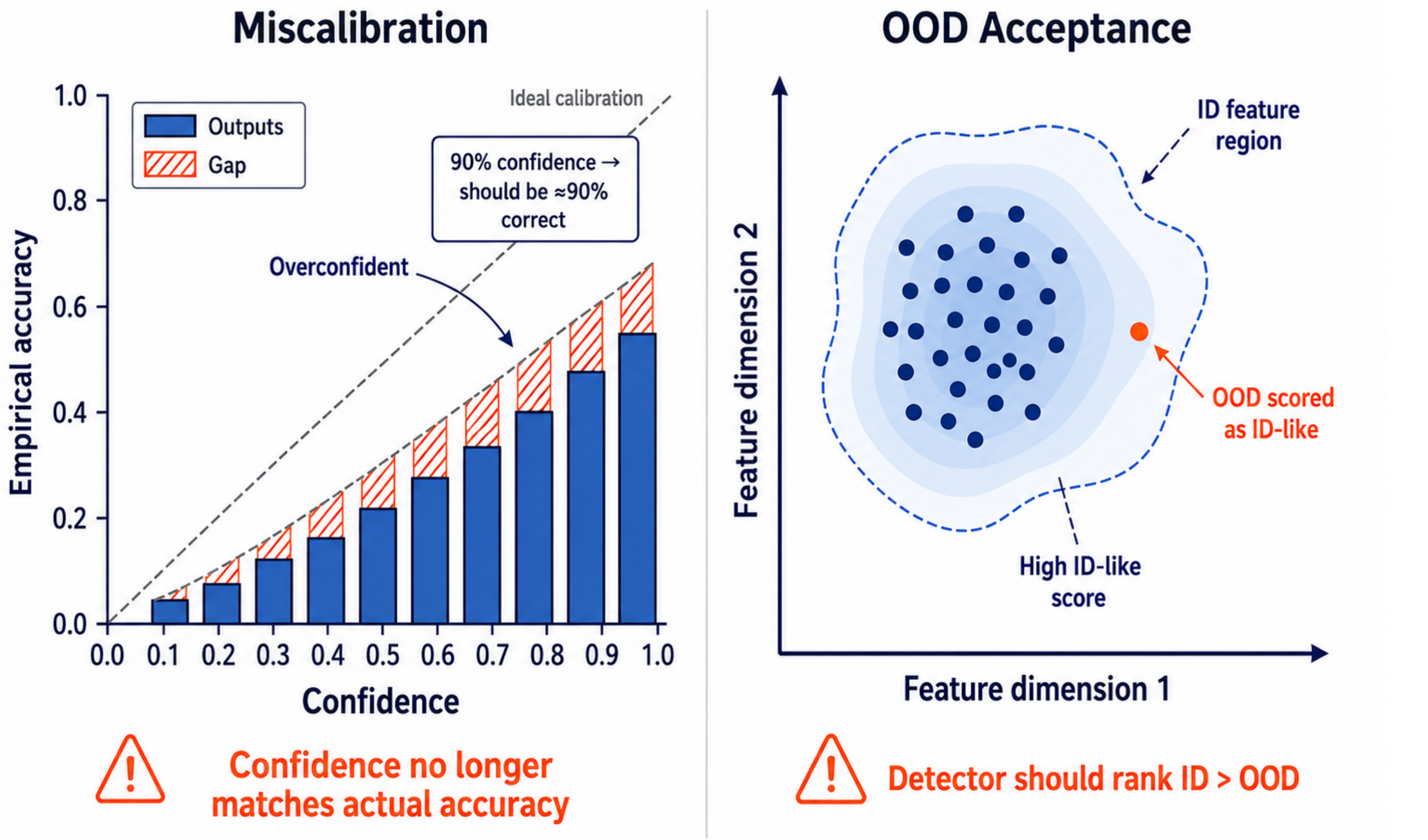


Figure 1. Left: confidence no longer matches empirical accuracy, destabilizing threshold decisions. Right: an OOD input receives a high ID-like score and can be passed downstream as if it were ID.

Metric definitions

Calibration

$$\text{acc}(B) = \frac{1}{|B|} \sum_{i \in B} \mathbf{1}(\hat{y}_i = y_i),$$

$$\text{conf}(B) = \frac{1}{|B|} \sum_{i \in B} \max_k p_i(k)$$

$$\text{ECE} = \sum_g \frac{|B_g|}{n} |\text{acc}(B_g) - \text{conf}(B_g)|$$

ECE summarizes the confidence-accuracy gap across confidence bins.

OOD ranking

$$s(x) : \text{ID-like score}$$

$$\text{AUROC} = P\{s(x_{\text{ID}}) > s(x_{\text{OOD}})\}$$

AUROC summarizes whether ID samples receive higher scores than OOD samples.

Optimizers: Update Rules Shift Feature Distributions

최적화 방법은 마지막 표현층의 class-conditional feature distribution을 바꿀 수 있다.

Feature distribution view

마지막 표현을 $H = h_\theta(X)$ 로 보면, optimizer는 각 class 조건부분포 $H \mid Y = k$ 의 위치와 산포를 만드는 implicit bias로 작용한다.

$$H \mid Y = k : \quad \mu_k = \mathbb{E}[H \mid Y = k], \quad \Sigma_k = \text{Var}(H \mid Y = k)$$

Feature detectors see within-class dispersion, covariance scale/anisotropy, and between-class separation.

Weight-decay coupling mechanism

$$z = W h_\theta(x), \quad s_t = W_t^\top \mathbf{1}_K, \quad a_t = \frac{1}{K} \|s_t\|_2^2, \quad (\nabla_W L_{\text{CE}})^\top \mathbf{1}_K = 0$$

SGD/L2: centering contraction

$$s_{t+1} = (1 - \eta\lambda)s_t$$

$$a_{t+1} \approx (1 - \eta\lambda)^2 a_t$$

CE gradient row-sum이 0이므로 class-direction centering statistic이 수축한다.

Adam/L2 vs AdamW: distribution path split

$$\text{Adam/L2: } D_t(\nabla L + \lambda W_t)$$

$$\text{AdamW: } D_t \nabla L + \text{outside shrinkage}$$

$$(D_t \nabla L)^\top \mathbf{1}_K \neq 0$$

좌표별 scaling이 classifier centering과 feature 분포 경로를 바꿀 수 있다.

Implication Accuracy가 비슷해도 class mean separation, within-class dispersion, covariance scale/anisotropy, classifier-feature alignment가 달라질 수 있다.

Boundary OOD reliability link is evaluated by our post-hoc diagnostics.

Experiment

Selection uses ID validation accuracy only.

ID / model CIFAR-10 / WRN-28-10

Factors SGD, Adam, AdamW × LR-WD grid

Selection ID validation accuracy only

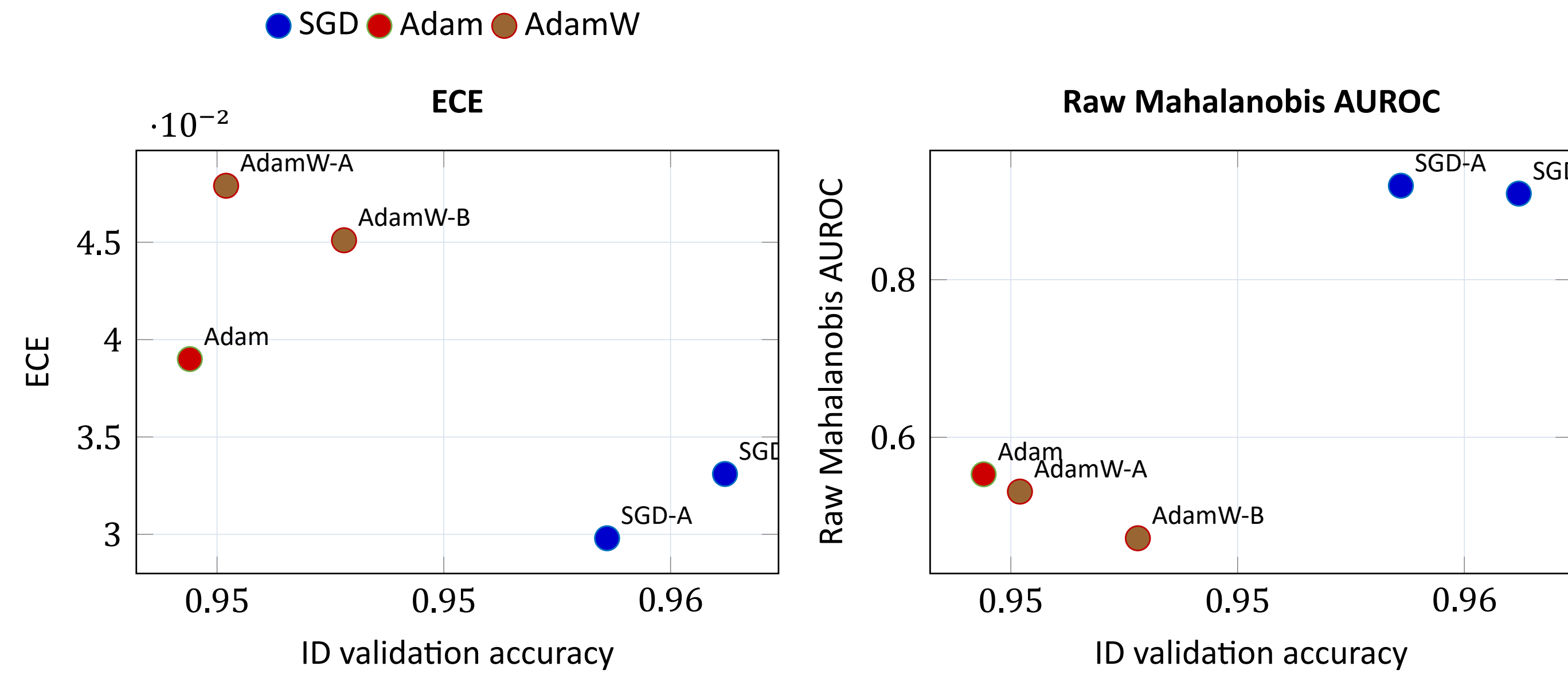
Diagnostics ECE, OOD AUROC, feature distribution metrics

OOD CIFAR-100, TinyImageNet, SVHN, MNIST

Status Seed0 diagnostic draft

Figure 2. Accuracy-Matched Reliability Split

Figure 2. Accuracy-Matched Reliability Scatter (seed0 draft)



Seed0 diagnostic draft: ECE and raw Mahalanobis AUROC against ID validation accuracy. Calibration mismatch and raw feature OOD reliability can split within a similar ID accuracy band.

Read Similar ID validation accuracy does not imply similar reliability.

Evidence Seed0 candidates in a competitive accuracy band split in ECE and raw Mahalanobis AUROC.

Boundary Diagnostic draft; repeated-seed summaries will replace this view.

Table 1. Dataset-Specific Raw Mahalanobis AUROC

Config	Optimizer	CIFAR-100	TinyImageNet	SVHN	MNIST
SGD-B	SGD	0.861	0.857	0.973	0.946
Adam-O	Adam	0.610	0.565	0.900	0.630
Adam	Adam	0.568	0.542	0.660	0.442
AdamW-B	AdamW	0.435	0.398	0.783	0.271

Dataset-specific raw Mahalanobis AUROC for four confirmed seed0 representative rows. Higher is better. Current table is a seed0 diagnostic draft and will be replaced by repeated-seed summaries.

Read Raw Mahalanobis AUROC differs by OOD dataset and config.

Evidence Confirmed seed0 representative rows show large dataset-specific drops for Adam/AdamW rows relative to SGD-B.

Boundary Four directly confirmed representative rows only; no seed-averaged optimizer ranking claim.

Table 2. Geometry Summary

Config	Opt	Val/Test	ECE	Raw Maha	NC1	InterDist	Geom. diag.
SGD-A	SGD	95.86/95.85	2.98	0.919	0.049	16.80	compact
SGD-B	SGD	96.12/95.46	3.31	0.909	0.068	14.33	compact
Adam	Adam	94.94/94.47	3.90	0.553	0.187	13.42	spread ↑
AdamW-B	AdamW	95.28/94.68	4.51	0.472	0.271	5.37	separation ↓
AdamW-A	AdamW	95.02/94.37	4.79	0.531	0.275	5.53	separation ↓

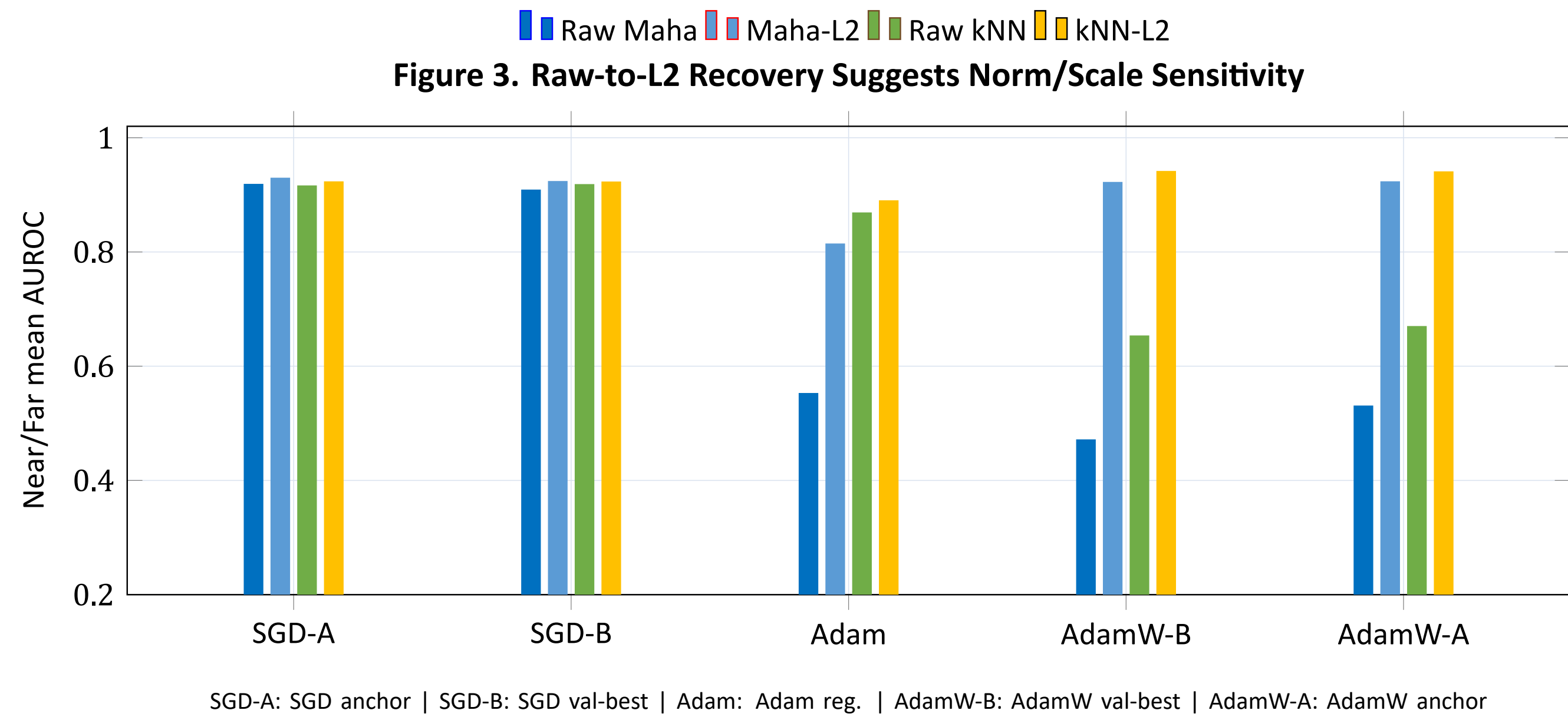
Geometry diagnostics for selected accuracy-matched configs. Lower NC1 and larger InterDist generally indicate more compact and separated class geometry. Additional norm/covariance diagnostics are included when confirmed.

Read Reliability split is accompanied by different penultimate geometry.

Evidence Higher NC1 indicates larger within-class spread relative to separation; smaller InterDist indicates weaker class-mean separation.

Implication Distance/covariance-based detectors can see a different score landscape even when ID accuracy is similar.

Figure 3. Raw-to-L2 Recovery Suggests Norm/Scale Sensitivity



Raw-to-L2 recovery suggests norm/covariance-scale sensitivity. L2-normalized controls reduce feature-scale effects while preserving angular information.

Read Raw-to-L2 recovery suggests norm/scale sensitivity.

Evidence L2-normalized controls reduce feature-scale effects while preserving angular information.

Boundary Diagnostic control only; not a claim that L2 normalization is the final detector solution.

Mechanism Diagnostic and Poster Conclusion

Detector reliability depends on the geometry it sees, not the optimizer name directly.

Mechanism Raw Mahalanobis uses class means and covariance-based distance; raw kNN uses neighborhood distance.

Sensitivity Both can change with feature norms, class separation, covariance anisotropy, and effective rank.

Conclusion Accuracy is necessary but not sufficient. ID-validation-matched models can show different calibration, raw feature OOD reliability, and feature geometry profiles.

Boundary Seed0 diagnostic evidence; not a causal proof or seed-averaged conclusion.

Future Work

- Adam-AdamW interpolation: coupled/decoupled weight decay 비율을 연속적으로 조정한다.
- Dataset / architecture expansion: CIFAR-10 WRN-28-10 밖에서도 패턴을 확인한다.
- Other training methods: SAM, Mixup 등이 feature norm/covariance scale과 detector reliability를 바꾸는지 확장한다.